

Electoral Judicialization and Municipal Competition in Brazil

Evidence from a Bartik Shift-Share Instrument

Nara Livia Moraes

FEA-USP

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Does Judicial Intervention Shape Electoral Competition?

Research question:

Does pre-election judicial intervention causally affect the competitive dynamics of Brazilian mayoral races?

The mechanism:

- Candidates face adversarial lawsuits *before* election day
- Suits impose legal costs, reputational exposure, and campaign disruption
- This could deter entry, shift vote shares, and alter voter behavior

The challenge:

Close races *attract* more lawsuits — endogeneity runs in both directions.

Strategy: Bartik shift-share IV predicting lawsuit exposure from a municipality's pre-determined 2020 topic mix $\times$ state-level topic growth ($F \approx 19.6$).

Scale

- 5,560 Brazilian municipalities
- 2020 and 2024 mayoral elections
- TSE administrative lawsuit data
- 2,187 electoral zone clusters

Key Findings

- Blank vote rate $\uparrow$ significantly (+1.3 pp, $p = 0.018$)
- Effect concentrated in open-seat municipalities (+3.0 pp)
- Candidate pool contracts directionally (-6.6% , $p = 0.092$)
- Competition outcomes: precise nulls

Three-tier Electoral Justice:

- 1 **Zona Eleitoral (ZE)** — first instance;
2,800+ nationwide
- 2 **Tribunal Regional Eleitoral (TRE)** — 27
state appellate courts
- 3 **Tribunal Superior Eleitoral (TSE)** —
national apex

Courts adjudicate **before** election day: eligibility challenges, campaign conduct, abuse of power, propaganda violations.

Treatment: $\Delta \log(1 + \text{adversarial lawsuits}_m)$,
2020→2024.

Data come exclusively from **first-instance (ZE) filings** —
the level where pre-election challenges are lodged.

Appeals at TRE/TSE are excluded.

Administrative vs. Adversarial

Type	Included?
Candidacy registration (RRC)	×
Coalition registration (DRAP)	×
Mandatory procedural classes	×
Eligibility challenges	✓
Abuse of power	✓
Campaign conduct	✓
Propaganda violations	✓
Document/information offenses	✓

Two-cycle design

2020 (Nov 15) → **shares + baseline controls**
2024 (Oct 6) → **treatment + outcomes**

Template: Ash, Morelli & Vannoni (JPE, forthcoming)

AMV estimate the causal effect of legislative output on U.S. state economic growth using a Bartik shift-share IV: pre-period topic shares of legislation $\times$ leave-out growth of each topic in other states.

Why follow this design?

In both settings, jurisdictions enter the period with a pre-existing topic mix and are then hit by supra-jurisdictional waves — policy diffusion across U.S. states in AMV; TRE jurisprudence and MPE enforcement priorities across Brazilian municipalities here. The instrument isolates the component of growth each jurisdiction inherits from these common shocks, weighted by prior exposure. Neither unit can anticipate which topics will grow.

Why are the shifts exogenous?

Helmke & Staton (2011): litigation waves in Latin America diffuse through *supra-municipal* forces — TRE jurisprudence, MPE enforcement priorities, and coordinated party legal strategies. No single municipality drives state-level topic growth.

The 2020 topic shares are pre-determined: a municipality's lawsuit portfolio reflects politics four years before 2024 outcomes materialize (GPS 2020; BHJ 2022).

Key difference from AMV

AMV's first stage is negative — low-share states borrow more from common shocks. Here it is positive: municipalities more exposed to a topic in 2020 pick up more of that topic's state-level growth. Different absorption mechanism, identical identification logic.

$$Z_m^B = \sum_{k \notin \mathcal{A}} s_{m,k,2020} \cdot g_{UF(m), k, -m}$$

$s_{m,k,2020}$	Share of topic k in municipality m 's adversarial lawsuits in 2020; re-normalized to $\sum_k s_{m,k,2020} = 1$
$g_{UF,k,-m}$	Leave-one-out log growth of topic k in state $UF(m)$, 2020→2024: $g = \log\left(\frac{T_{UF,k,2024} - C_{m,k,2024}}{T_{UF,k,2020} - C_{m,k,2020}}\right)$
$\mathcal{A}$	Excluded administrative classes and subjects (RRC, DRAP, mandatory codes) applied at the data construction stage

Single preferred specification. One instrument ($F \approx 19.6$) used throughout. The adversarial filter is applied in the build pipeline — not post-hoc. This conservatively isolates genuine judicial intervention from administrative court activity.

First stage (OLS):

$$\underbrace{\Delta \log(1 + \ell_m)}_{\text{endogenous}} = \alpha + \beta Z_m^B + \mathbf{X}_m' \gamma + \delta_{\text{UF}(m)} + \varepsilon_m$$

Second stage (2SLS):

$$\Delta Y_m = \alpha + \tau \Delta \log(\widehat{1 + \ell_m}) + \mathbf{X}_m' \gamma + \delta_{\text{UF}(m)} + u_m$$

Specification

- δ_{UF} : state fixed effects (27 units) — absorbs common state shocks
- $\mathbf{X}_m$: 7 baseline controls (Census 2010 demographics, 2016 margin, 2020 electoral baseline)
- SE clustered by principal electoral zone (2,187 clusters)
- Δ = first difference 2024–2020

Outcome families

Electoral competition: runner-up share, winner–runner-up margin, winner share, winner majority, log candidates

Voter behavior: turnout rate, null vote rate, blank vote rate

Composition: female/nonwhite vote shares, winner identity

Specification	Coef.	SE	Ex. F	Lg. F
Baseline (state FE)	1.791	0.405	19.6	19.4
Single-zone	1.801	0.422	18.2	18.1
Full controls	1.800	0.406	19.7	19.4
$\leq 200k$ voters	1.869	0.426	19.3	19.2
Broader lawsuits ctrl	1.792	0.405	19.6	19.4

$N = 5,560$ baseline; 5,371 single-zone; 5,338/5,509 exec/leg $\leq 200k$.
Coef. $\hat{\beta} \approx 1.8$ and $F > 18$ across all specs and both arenas.

Instrument strength

$F = 19.6$ across all specifications. Well above the Stock–Yogo 10% size distortion threshold (≈ 10). Coefficient $\hat{\beta} \approx 1.8$ is stable across all robustness checks.

LIML check

For a just-identified model ($K = 1$),
 $\text{LIML} \equiv 2\text{SLS}$ exactly. Max divergence = 0.0000 across all outcomes.

Rotemberg decomposition

Top-2 topics by weight:

- Document falsification ($\alpha = +24\%$, $F_k = 51$)
- Fraudulent electoral polls ($\alpha = -23\%$, $F_k = 60$)

Both have strong per-topic first stages.

Electoral Competition: Executive vs. Legislative

Executive (mayoral) — $N = 5,560$, $F = 19.6$

Baseline: state FE + 7 controls

Outcome	Coef.	SE	p
Δ Runner-up share	-0.031	0.022	0.157
Δ Margin (W-RU)	+0.062	0.043	0.151
Δ Winner share	+0.031	0.025	0.213
Δ Winner majority	+0.140	0.090	0.122
Δ Log candidates	-0.066	0.039	0.092

Legislative (council) — $N = 5,560$, $F = 19.4$

Baseline: state FE + 7 controls

Outcome	Coef.	SE	p
Δ Log candidates	-0.075	0.041	0.066
Δ Female share	-0.006	0.006	0.370
Δ Nonwhite share	-0.018	0.019	0.346
Δ New cand. share	-0.020	0.016	0.217
Δ Incumb. share	-0.008	0.010	0.415

Robustness: $\leq 200k$ municipalities $\rightarrow$ log candidates -0.079

($p = 0.058$).

Entry deterrence replicates; competition margins remain imprecise

Fewer candidates (-6.6% executive, -7.5% legislative) replicates across both arenas and all specifications. Restricting to $\leq 200k$ -voter municipalities — where legal resources and party machines are thinner — brings legislative $p = 0.058$. Competition margins (runner-up share, winner majority) move in the expected direction but cannot be pinned down statistically.

Composition: Executive vs. Legislative

$N = 5,560$. Baseline: state FE + 7 controls. $F_{exec} = 19.6$, $F_{leg} = 19.4$.

Executive: vote share going to each group. Legislative: share of elected council members from each group.

Outcome	Executive (mayoral)			Legislative (council)		
	Coef.	SE	p	Coef.	SE	p
Δ Female share	-0.093	0.048	0.052	+0.013	0.022	0.554
Δ Nonwhite share	-0.039	0.042	0.348	-0.027	0.026	0.288
Δ Incumbent share	+0.010	0.049	0.844	+0.006	0.025	0.797
Δ Winner female	-0.123	0.063	0.052	<i>(multi-winner race)</i>		

Female representation: mayoral effect, council null

Adversarial litigation reduces female candidates' vote share and the probability of a female winner in the mayoral race ($p \approx 0.052$ for both). The city council shows precise nulls. The contrast likely reflects race structure: a single-winner mayoral race has one identifiable target; council races diffuse litigation across large multicandidate pools.

Results: Voter Behavior and Open-Seat Heterogeneity

Outcome	Coef.	SE	<i>p</i>
Δ Turnout rate	-0.006	0.005	0.185
Δ Null vote rate	+0.007	0.006	0.220
Δ Blank vote rate	+0.013	0.005	0.018

Blank vote rate: the main finding.

At mean 1.9%, +1.3 pp is a $\approx 10\%$ increase ($p = 0.018$).

Open-seat heterogeneity (blank rate):

Sample	Coef.	SE	<i>p</i>
All ($N = 5,560$)	+0.013	0.005	0.018
Open seat ($N = 2,571$)	+0.030	0.014	0.028
Contested ($N = 2,989$)	+0.003	0.005	0.546

Open seat $F = 10.6$; contested seat $F = 15.3$.

Mechanism: judicial ambiguity

Adversarial challenges are public records. Intense litigation generates *diffuse* reputational noise across all candidates, not targeted signals. Voters respond with expressive abstention (blank vote), not non-participation (turnout null).

Why open seats?

Open seat (46%): 2020 winner served two consecutive terms, constitutionally term-limited. No focal incumbent. Judicial noise creates ambiguity about *all* contenders, elevating blank vote.

Contested seat (54%): incumbent provides a valence benchmark; legal noise is absorbed into the for/against incumbent calculus.

What seems to be happening:

- The clearest result is the blank vote (+1.3 pp, $p = 0.018$). I read this as voters becoming unsure about everyone, not reacting to a specific candidate
- What surprised me: the effect is almost entirely in open seats and disappears when incumbents run. Maybe voters need a focal candidate to absorb the noise
- I consistently find fewer candidates (−6–7%) across both arenas, but pool composition does not shift. More like raising a cost than filtering by type
- Competition outcomes all go in the expected direction, but I cannot pin them down statistically
- Turnout does not move. Judicial activity does not demobilize voters, which I think matters

Next steps

- **Pre-period placebo (2016→2020):** key robustness test. Pre-2018 TSE data not publicly available; formal request submitted to TSE under the *Lei de Acesso à Informação*. Awaiting response.
- **Candidate-level:** gender/race heterogeneity requires matching lawsuit respondents to candidate registry
- **TRE judge composition:** court composition as moderator of enforcement intensity
- **Case outcomes:** ruling win/loss to separate “filed” from “won” effects